Tiny Changes,
Remarkable Results

Atomic Habits

An Easy & Proven Way to
Build Good Habits & Break Bad Ones

James Clear



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AVERY
an imprint of Penguin Random House
New York



AN IMPRINT OF PENGUIN RANDOM HOUSE LLC
375 Hudson Street
New York, New York 10014



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Ebook ISBN 9780735211308

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a·tom·ic

əˈtämik

1. an extremely small amount of a thing; the single irreducible unit of a larger system.
2. the source of immense energy or power.

hab·it

ˈhabət

1. a routine or practice performed regularly; an automatic response to a specific situation.

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Introduction

My Story

On the final day of my sophomore year of high school, I was hit in the face with a baseball bat. As my classmate took a full swing, the bat slipped out of his hands and came flying toward me before striking me directly between the eyes. I have no memory of the moment of impact.

The bat smashed into my face with such force that it crushed my nose into a distorted U-shape. The collision sent the soft tissue of my brain slamming into the inside of my skull. Immediately, a wave of swelling surged throughout my head. In a fraction of a second, I had a broken nose, multiple skull fractures, and two shattered eye sockets.

When I opened my eyes, I saw people staring at me and running over to help. I looked down and noticed spots of red on my clothes. One of my classmates took the shirt off his back and handed it to me. I used it to plug the stream of blood rushing from my broken nose. Shocked and confused, I was unaware of how seriously I had been injured.

My teacher looped his arm around my shoulder and we began the long walk to the nurse's office: across the field, down the hill, and back into school. Random hands touched my sides, holding me upright. We took our time and walked slowly. Nobody realized that every minute mattered.

When we arrived at the nurse's office, she asked me a series of questions.

"What year is it?"

"1998," I answered. It was actually 2002.

"Who is the president of the United States?"

"Bill Clinton," I said. The correct answer was George W. Bush.

"What is your mom's name?"

"Uh. Um." I stalled. Ten seconds passed.

"Patti," I said casually, ignoring the fact that it had taken me ten seconds to remember my own mother's name.

That is the last question I remember. My body was unable to handle the rapid swelling in my brain and I lost consciousness before the ambulance arrived. Minutes later, I was carried out of school and taken to the local hospital.

Shortly after arriving, my body began shutting down. I struggled with basic functions like swallowing and breathing. I had my first seizure of the day. Then I

stopped breathing entirely. As the doctors hurried to supply me with oxygen, they also decided the local hospital was unequipped to handle the situation and ordered a helicopter to fly me to a larger hospital in Cincinnati.

I was rolled out of the emergency room doors and toward the helipad across the street. The stretcher rattled on a bumpy sidewalk as one nurse pushed me along while another pumped each breath into me by hand. My mother, who had arrived at the hospital a few moments before, climbed into the helicopter beside me. I remained unconscious and unable to breathe on my own as she held my hand during the flight.

While my mother rode with me in the helicopter, my father went home to check on my brother and sister and break the news to them. He choked back tears as he explained to my sister that he would miss her eighth-grade graduation ceremony that night. After passing my siblings off to family and friends, he drove to Cincinnati to meet my mother.

When my mom and I landed on the roof of the hospital, a team of nearly twenty doctors and nurses sprinted onto the helipad and wheeled me into the trauma unit. By this time, the swelling in my brain had become so severe that I was having repeated post-traumatic seizures. My broken bones needed to be fixed, but I was in no condition to undergo surgery. After yet another seizure—my third of the day—I was put into a medically induced coma and placed on a ventilator.

My parents were no strangers to this hospital. Ten years earlier, they had entered the same building on the ground floor after my sister was diagnosed with leukemia at age three. I was five at the time. My brother was just six months old. After two and a half years of chemotherapy treatments, spinal taps, and bone marrow biopsies, my little sister finally walked out of the hospital happy, healthy, and cancer free. And now, after ten years of normal life, my parents found themselves back in the same place with a different child.

While I slipped into a coma, the hospital sent a priest and a social worker to comfort my parents. It was the same priest who had met with them a decade earlier on the evening they found out my sister had cancer.

As day faded into night, a series of machines kept me alive. My parents slept restlessly on a hospital mattress—one moment they would collapse from fatigue, the next they would be wide awake with worry. My mother would tell me later, “It was one of the worst nights I’ve ever had.”

MY RECOVERY

Mercifully, by the next morning my breathing had rebounded to the point where the doctors felt comfortable releasing me from the coma. When I finally regained consciousness, I discovered that I had lost my ability to smell. As a test, a nurse asked me to blow my nose and sniff an apple juice box. My sense of smell returned, but—to everyone's surprise—the act of blowing my nose forced air through the fractures in my eye socket and pushed my left eye outward. My eyeball bulged out of the socket, held precariously in place by my eyelid and the optic nerve attaching my eye to my brain.

The ophthalmologist said my eye would gradually slide back into place as the air seeped out, but it was hard to tell how long this would take. I was scheduled for surgery one week later, which would allow me some additional time to heal. I looked like I had been on the wrong end of a boxing match, but I was cleared to leave the hospital. I returned home with a broken nose, half a dozen facial fractures, and a bulging left eye.

The following months were hard. It felt like everything in my life was on pause. I had double vision for weeks; I literally couldn't see straight. It took more than a month, but my eyeball did eventually return to its normal location. Between the seizures and my vision problems, it was eight months before I could drive a car again. At physical therapy, I practiced basic motor patterns like walking in a straight line. I was determined not to let my injury get me down, but there were more than a few moments when I felt depressed and overwhelmed.

I became painfully aware of how far I had to go when I returned to the baseball field one year later. Baseball had always been a major part of my life. My dad had played minor league baseball for the St. Louis Cardinals, and I had a dream of playing professionally, too. After months of rehabilitation, what I wanted more than anything was to get back on the field.

But my return to baseball was not smooth. When the season rolled around, I was the only junior to be cut from the varsity baseball team. I was sent down to play with the sophomores on junior varsity. I had been playing since age four, and for someone who had spent so much time and effort on the sport, getting cut was humiliating. I vividly remember the day it happened. I sat in my car and cried as I flipped through the radio, desperately searching for a song that would make me feel better.

After a year of self-doubt, I managed to make the varsity team as a senior, but I rarely made it on the field. In total, I played eleven innings of high school varsity baseball, barely more than a single game.

Despite my lackluster high school career, I still believed I could become a great player. And I knew that if things were going to improve, I was the one responsible for making it happen. The turning point came two years after my

injury, when I began college at Denison University. It was a new beginning, and it was the place where I would discover the surprising power of small habits for the first time.

HOW I LEARNED ABOUT HABITS

Attending Denison was one of the best decisions of my life. I earned a spot on the baseball team and, although I was at the bottom of the roster as a freshman, I was thrilled. Despite the chaos of my high school years, I had managed to become a college athlete.

I wasn't going to be starting on the baseball team anytime soon, so I focused on getting my life in order. While my peers stayed up late and played video games, I built good sleep habits and went to bed early each night. In the messy world of a college dorm, I made a point to keep my room neat and tidy. These improvements were minor, but they gave me a sense of control over my life. I started to feel confident again. And this growing belief in myself rippled into the classroom as I improved my study habits and managed to earn straight A's during my first year.

A habit is a routine or behavior that is performed regularly—and, in many cases, automatically. As each semester passed, I accumulated small but consistent habits that ultimately led to results that were unimaginable to me when I started. For example, for the first time in my life, I made it a habit to lift weights multiple times per week, and in the years that followed, my six-foot-four-inch frame bulked up from a featherweight 170 to a lean 200 pounds.

When my sophomore season arrived, I earned a starting role on the pitching staff. By my junior year, I was voted team captain and at the end of the season, I was selected for the all-conference team. But it was not until my senior season that my sleep habits, study habits, and strength-training habits really began to pay off.

Six years after I had been hit in the face with a baseball bat, flown to the hospital, and placed into a coma, I was selected as the top male athlete at Denison University and named to the ESPN Academic All-America Team—an honor given to just thirty-three players across the country. By the time I graduated, I was listed in the school record books in eight different categories. That same year, I was awarded the university's highest academic honor, the President's Medal.

I hope you'll forgive me if this sounds boastful. To be honest, there was nothing legendary or historic about my athletic career. I never ended up playing

professionally. However, looking back on those years, I believe I accomplished something just as rare: I fulfilled my potential. And I believe the concepts in this book can help you fulfill your potential as well.

We all face challenges in life. This injury was one of mine, and the experience taught me a critical lesson: changes that seem small and unimportant at first will compound into remarkable results if you're willing to stick with them for years. We all deal with setbacks but in the long run, the quality of our lives often depends on the quality of our habits. With the same habits, you'll end up with the same results. But with better habits, anything is possible.

Maybe there are people who can achieve incredible success overnight. I don't know any of them, and I'm certainly not one of them. There wasn't one defining moment on my journey from medically induced coma to Academic All-American; there were many. It was a gradual evolution, a long series of small wins and tiny breakthroughs. The only way I made progress—the only choice I had—was to start small. And I employed this same strategy a few years later when I started my own business and began working on this book.

HOW AND WHY I WROTE THIS BOOK

In November 2012, I began publishing articles at jamesclear.com. For years, I had been keeping notes about my personal experiments with habits and I was finally ready to share some of them publicly. I began by publishing a new article every Monday and Thursday. Within a few months, this simple writing habit led to my first one thousand email subscribers, and by the end of 2013 that number had grown to more than thirty thousand people.

In 2014, my email list expanded to over one hundred thousand subscribers, which made it one of the fastest-growing newsletters on the internet. I had felt like an impostor when I began writing two years earlier, but now I was becoming known as an expert on habits—a new label that excited me but also felt uncomfortable. I had never considered myself a master of the topic, but rather someone who was experimenting alongside my readers.

In 2015, I reached two hundred thousand email subscribers and signed a book deal with Penguin Random House to begin writing the book you are reading now. As my audience grew, so did my business opportunities. I was increasingly asked to speak at top companies about the science of habit formation, behavior change, and continuous improvement. I found myself delivering keynote speeches at conferences in the United States and Europe.

In 2016, my articles began to appear regularly in major publications like *Time*,

Entrepreneur, and *Forbes*. Incredibly, my writing was read by over eight million people that year. Coaches in the NFL, NBA, and MLB began reading my work and sharing it with their teams.

At the start of 2017, I launched the Habits Academy, which became the premier training platform for organizations and individuals interested in building better habits in life and work.* Fortune 500 companies and growing start-ups began to enroll their leaders and train their staff. In total, over ten thousand leaders, managers, coaches, and teachers have graduated from the Habits Academy, and my work with them has taught me an incredible amount about what it takes to make habits work in the real world.

As I put the finishing touches on this book in 2018, jamesclear.com is receiving millions of visitors per month and nearly five hundred thousand people subscribe to my weekly email newsletter—a number that is so far beyond my expectations when I began that I'm not even sure what to think of it.

HOW THIS BOOK WILL BENEFIT YOU

The entrepreneur and investor Naval Ravikant has said, "To write a great book, you must first become the book." I originally learned about the ideas mentioned here because I had to live them. I had to rely on small habits to rebound from my injury, to get stronger in the gym, to perform at a high level on the field, to become a writer, to build a successful business, and simply to develop into a responsible adult. Small habits helped me fulfill my potential, and since you picked up this book, I'm guessing you'd like to fulfill yours as well.

In the pages that follow, I will share a step-by-step plan for building better habits—not for days or weeks, but for a lifetime. While science supports everything I've written, this book is not an academic research paper; it's an operating manual. You'll find wisdom and practical advice front and center as I explain the science of how to create and change your habits in a way that is easy to understand and apply.

The fields I draw on—biology, neuroscience, philosophy, psychology, and more—have been around for many years. What I offer you is a synthesis of the best ideas smart people figured out a long time ago as well as the most compelling discoveries scientists have made recently. My contribution, I hope, is to find the ideas that matter most and connect them in a way that is highly actionable. Anything wise in these pages you should credit to the many experts who preceded me. Anything foolish, assume it is my error.

The backbone of this book is my four-step model of habits—cue, craving,

response, and reward—and the four laws of behavior change that evolve out of these steps. Readers with a psychology background may recognize some of these terms from operant conditioning, which was first proposed as "stimulus, response, reward" by B. F. Skinner in the 1930s and has been popularized more recently as "cue, routine, reward" in *The Power of Habit* by Charles Duhigg.

Behavioral scientists like Skinner realized that if you offered the right reward or punishment, you could get people to act in a certain way. But while Skinner's model did an excellent job of explaining how external stimuli influenced our habits, it lacked a good explanation for how our thoughts, feelings, and beliefs impact our behavior. Internal states—our moods and emotions—matter, too. In recent decades, scientists have begun to determine the connection between our thoughts, feelings, and behavior. This research will also be covered in these pages.

In total, the framework I offer is an integrated model of the cognitive and behavioral sciences. I believe it is one of the first models of human behavior to accurately account for both the influence of external stimuli and internal emotions on our habits. While some of the language may be familiar, I am confident that the details—and the applications of the Four Laws of Behavior Change—will offer a new way to think about your habits.

Human behavior is always changing: situation to situation, moment to moment, second to second. But this book is about what *doesn't* change. It's about the fundamentals of human behavior. The lasting principles you can rely on year after year. The ideas you can build a business around, build a family around, build a life around.

There is no one right way to create better habits, but this book describes the best way I know—an approach that will be effective regardless of where you start or what you're trying to change. The strategies I cover will be relevant to anyone looking for a step-by-step system for improvement, whether your goals center on health, money, productivity, relationships, or all of the above. As long as human behavior is involved, this book will be your guide.

THE FUNDAMENTALS

Why Tiny Changes Make a Big Difference



1

The Surprising Power of Atomic Habits

The fate of British Cycling changed one day in 2003. The organization, which was the governing body for professional cycling in Great Britain, had recently hired Dave Brailsford as its new performance director. At the time, professional cyclists in Great Britain had endured nearly one hundred years of mediocrity. Since 1908, British riders had won just a single gold medal at the Olympic Games, and they had fared even worse in cycling's biggest race, the Tour de France. In 110 years, no British cyclist had ever won the event.

In fact, the performance of British riders had been so underwhelming that one of the top bike manufacturers in Europe refused to sell bikes to the team because they were afraid that it would hurt sales if other professionals saw the Brits using their gear.

Brailsford had been hired to put British Cycling on a new trajectory. What made him different from previous coaches was his relentless commitment to a strategy that he referred to as "the aggregation of marginal gains," which was the philosophy of searching for a tiny margin of improvement in everything you do. Brailsford said, "The whole principle came from the idea that if you broke down everything you could think of that goes into riding a bike, and then improve it by 1 percent, you will get a significant increase when you put them all together."

Brailsford and his coaches began by making small adjustments you might expect from a professional cycling team. They redesigned the bike seats to make them more comfortable and rubbed alcohol on the tires for a better grip. They asked riders to wear electrically heated overshorts to maintain ideal muscle temperature while riding and used biofeedback sensors to monitor how each athlete responded to a particular workout. The team tested various fabrics in a wind tunnel and had their outdoor riders switch to indoor racing suits, which proved to be lighter and more aerodynamic.

But they didn't stop there. Brailsford and his team continued to find 1 percent

improvements in overlooked and unexpected areas. They tested different types of massage gels to see which one led to the fastest muscle recovery. They hired a surgeon to teach each rider the best way to wash their hands to reduce the chances of catching a cold. They determined the type of pillow and mattress that led to the best night's sleep for each rider. They even painted the inside of the team truck white, which helped them spot little bits of dust that would normally slip by unnoticed but could degrade the performance of the finely tuned bikes.

As these and hundreds of other small improvements accumulated, the results came faster than anyone could have imagined.

Just five years after Brailsford took over, the British Cycling team dominated the road and track cycling events at the 2008 Olympic Games in Beijing, where they won an astounding 60 percent of the gold medals available. Four years later, when the Olympic Games came to London, the Brits raised the bar as they set nine Olympic records and seven world records.

That same year, Bradley Wiggins became the first British cyclist to win the Tour de France. The next year, his teammate Chris Froome won the race, and he would go on to win again in 2015, 2016, and 2017, giving the British team five Tour de France victories in six years.

During the ten-year span from 2007 to 2017, British cyclists won 178 world championships and sixty-six Olympic or Paralympic gold medals and captured five Tour de France victories in what is widely regarded as the most successful run in cycling history.*

How does this happen? How does a team of previously ordinary athletes transform into world champions with tiny changes that, at first glance, would seem to make a modest difference at best? Why do small improvements accumulate into such remarkable results, and how can you replicate this approach in your own life?

WHY SMALL HABITS MAKE A BIG DIFFERENCE

It is so easy to overestimate the importance of one defining moment and underestimate the value of making small improvements on a daily basis. Too often, we convince ourselves that massive success requires massive action. Whether it is losing weight, building a business, writing a book, winning a championship, or achieving any other goal, we put pressure on ourselves to make some earth-shattering improvement that everyone will talk about.

Meanwhile, improving by 1 percent isn't particularly notable—sometimes it isn't even *noticeable*—but it can be far more meaningful, especially in the long

run. The difference a tiny improvement can make over time is astounding. Here's how the math works out: if you can get 1 percent better each day for one year, you'll end up thirty-seven times better by the time you're done. Conversely, if you get 1 percent worse each day for one year, you'll decline nearly down to zero. What starts as a small win or a minor setback accumulates into something much more.

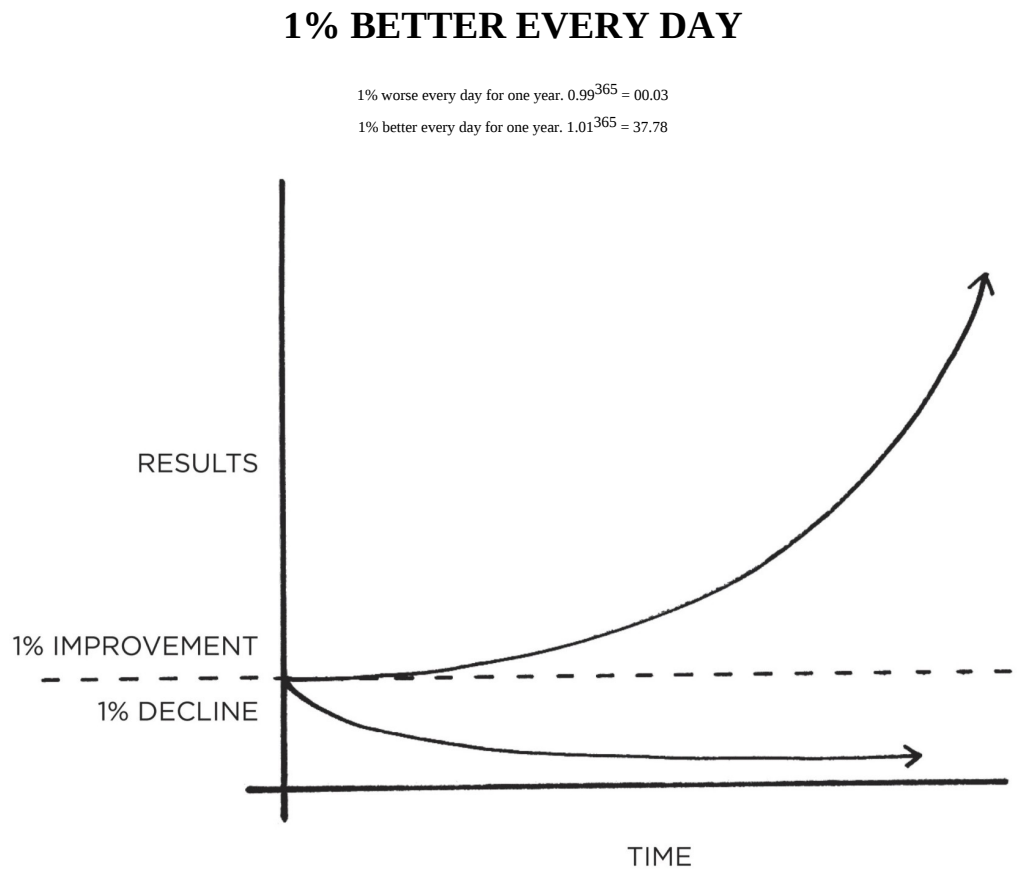


FIGURE 1: The effects of small habits compound over time. For example, if you can get just 1 percent better each day, you'll end up with results that are nearly 37 times better after one year.

Habits are the compound interest of selfimprovement. The same way that money multiplies through compound interest, the effects of your habits multiply as you repeat them. They seem to make little difference on any given day and yet the impact they deliver over the months and years can be enormous. It is only when looking back two, five, or perhaps ten years later that the value of good habits and the cost of bad ones becomes strikingly apparent.

This can be a difficult concept to appreciate in daily life. We often dismiss small changes because they don't seem to matter very much in the moment. If

you save a little money now, you're still not a millionaire. If you go to the gym three days in a row, you're still out of shape. If you study Mandarin for an hour tonight, you still haven't learned the language. We make a few changes, but the results never seem to come quickly and so we slide back into our previous routines.

Unfortunately, the slow pace of transformation also makes it easy to let a bad habit slide. If you eat an unhealthy meal today, the scale doesn't move much. If you work late tonight and ignore your family, they will forgive you. If you procrastinate and put your project off until tomorrow, there will usually be time to finish it later. A single decision is easy to dismiss.

But when we repeat 1 percent errors, day after day, by replicating poor decisions, duplicating tiny mistakes, and rationalizing little excuses, our small choices compound into toxic results. It's the accumulation of many missteps—a 1 percent decline here and there—that eventually leads to a problem.

The impact created by a change in your habits is similar to the effect of shifting the route of an airplane by just a few degrees. Imagine you are flying from Los Angeles to New York City. If a pilot leaving from LAX adjusts the heading just 3.5 degrees south, you will land in Washington, D.C., instead of New York. Such a small change is barely noticeable at takeoff—the nose of the airplane moves just a few feet—but when magnified across the entire United States, you end up hundreds of miles apart.*

Similarly, a slight change in your daily habits can guide your life to a very different destination. Making a choice that is 1 percent better or 1 percent worse seems insignificant in the moment, but over the span of moments that make up a lifetime these choices determine the difference between who you are and who you could be. Success is the product of daily habits—not once-in-a-lifetime transformations.

That said, it doesn't matter how successful or unsuccessful you are right now. What matters is whether your habits are putting you on the path toward success. You should be far more concerned with your current trajectory than with your current results. If you're a millionaire but you spend more than you earn each month, then you're on a bad trajectory. If your spending habits don't change, it's not going to end well. Conversely, if you're broke, but you save a little bit every month, then you're on the path toward financial freedom—even if you're moving slower than you'd like.

Your outcomes are a lagging measure of your habits. Your net worth is a lagging measure of your financial habits. Your weight is a lagging measure of your eating habits. Your knowledge is a lagging measure of your learning habits. Your clutter is a lagging measure of your cleaning habits. You get what you

repeat.

If you want to predict where you'll end up in life, all you have to do is follow the curve of tiny gains or tiny losses, and see how your daily choices will compound ten or twenty years down the line. Are you spending less than you earn each month? Are you making it into the gym each week? Are you reading books and learning something new each day? Tiny battles like these are the ones that will define your future self.

Time magnifies the margin between success and failure. It will multiply whatever you feed it. Good habits make time your ally. Bad habits make time your enemy.

Habits are a double-edged sword. Bad habits can cut you down just as easily as good habits can build you up, which is why understanding the details is crucial. You need to know how habits work and how to design them to your liking, so you can avoid the dangerous half of the blade.

YOUR HABITS CAN COMPOUND FOR YOU OR AGAINST YOU

Positive Compounding

Productivity compounds. Accomplishing one extra task is a small feat on any given day, but it counts for a lot over an entire career. The effect of automating an old task or mastering a new skill can be even greater. The more tasks you can handle without thinking, the more your brain is free to focus on other areas.

Knowledge compounds. Learning one new idea won't make you a genius, but a commitment to lifelong learning can be transformative. Furthermore, each book you read not only teaches you something new but also opens up different ways of thinking about old ideas. As Warren Buffett says, "That's how knowledge works. It builds up, like compound interest."

Relationships compound. People reflect your behavior back to you. The more you help others, the more others want to help you. Being a little bit nicer in each interaction can result in a network of broad and strong connections over time.

Negative Compounding

Stress compounds. The frustration of a traffic jam. The weight of parenting responsibilities. The worry of making ends meet. The strain of slightly high blood pressure. By themselves, these common causes of stress are manageable. But when they persist for years, little stresses compound into serious health issues.

Negative thoughts compound. The more you think of yourself as worthless, stupid, or ugly, the more you condition yourself to interpret life that way. You get trapped in a thought loop. The same is true for how you think about others. Once you fall into the habit of seeing people as angry, unjust, or selfish, you see those kind of people everywhere.

Outrage compounds. Riots, protests, and mass movements are rarely the result of a single event. Instead, a long series of microaggressions and daily aggravations slowly multiply until one event tips the scales and outrage spreads like wildfire.

WHAT PROGRESS IS REALLY LIKE

Imagine that you have an ice cube sitting on the table in front of you. The room is cold and you can see your breath. It is currently twenty-five degrees. Ever so slowly, the room begins to heat up.

Twenty-six degrees.

Twenty-seven.

Twenty-eight.

The ice cube is still sitting on the table in front of you.

Twenty-nine degrees.

Thirty.

Thirty-one.

Still, nothing has happened.

Then, thirty-two degrees. The ice begins to melt. A one-degree shift, seemingly no different from the temperature increases before it, has unlocked a huge change.

Breakthrough moments are often the result of many previous actions, which build up the potential required to unleash a major change. This pattern shows up everywhere. Cancer spends 80 percent of its life undetectable, then takes over the body in months. Bamboo can barely be seen for the first five years as it builds extensive root systems underground before exploding ninety feet into the air within six weeks.

Similarly, habits often appear to make no difference until you cross a critical threshold and unlock a new level of performance. In the early and middle stages of any quest, there is often a Valley of Disappointment. You expect to make progress in a linear fashion and it's frustrating how ineffective changes can seem during the first days, weeks, and even months. It doesn't feel like you are going anywhere. It's a hallmark of any compounding process: the most powerful outcomes are delayed.

This is one of the core reasons why it is so hard to build habits that last. People make a few small changes, fail to see a tangible result, and decide to stop. You think, "I've been running every day for a month, so why can't I see any change in my body?" Once this kind of thinking takes over, it's easy to let good habits fall by the wayside. But in order to make a meaningful difference, habits need to persist long enough to break through this plateau—what I call the *Plateau of Latent Potential*.

If you find yourself struggling to build a good habit or break a bad one, it is not because you have lost your ability to improve. It is often because you have not yet crossed the Plateau of Latent Potential. Complaining about not achieving success despite working hard is like complaining about an ice cube not melting when you heated it from twenty-five to thirty-one degrees. Your work was not wasted; it is just being stored. All the action happens at thirty-two degrees.

When you finally break through the Plateau of Latent Potential, people will call it an overnight success. The outside world only sees the most dramatic event rather than all that preceded it. But you know that it's the work you did long ago —when it seemed that you weren't making any progress—that makes the jump today possible.

It is the human equivalent of geological pressure. Two tectonic plates can grind against one another for millions of years, the tension slowly building all

the while. Then, one day, they rub each other once again, in the same fashion they have for ages, but this time the tension is too great. An earthquake erupts. Change can take years—before it happens all at once.

Mastery requires patience. The San Antonio Spurs, one of the most successful teams in NBA history, have a quote from social reformer Jacob Riis hanging in their locker room: “When nothing seems to help, I go and look at a stonecutter hammering away at his rock, perhaps a hundred times without as much as a crack showing in it. Yet at the hundred and first blow it will split in two, and I know it was not that last blow that did it—but all that had gone before.”

THE PLATEAU OF LATENT POTENTIAL

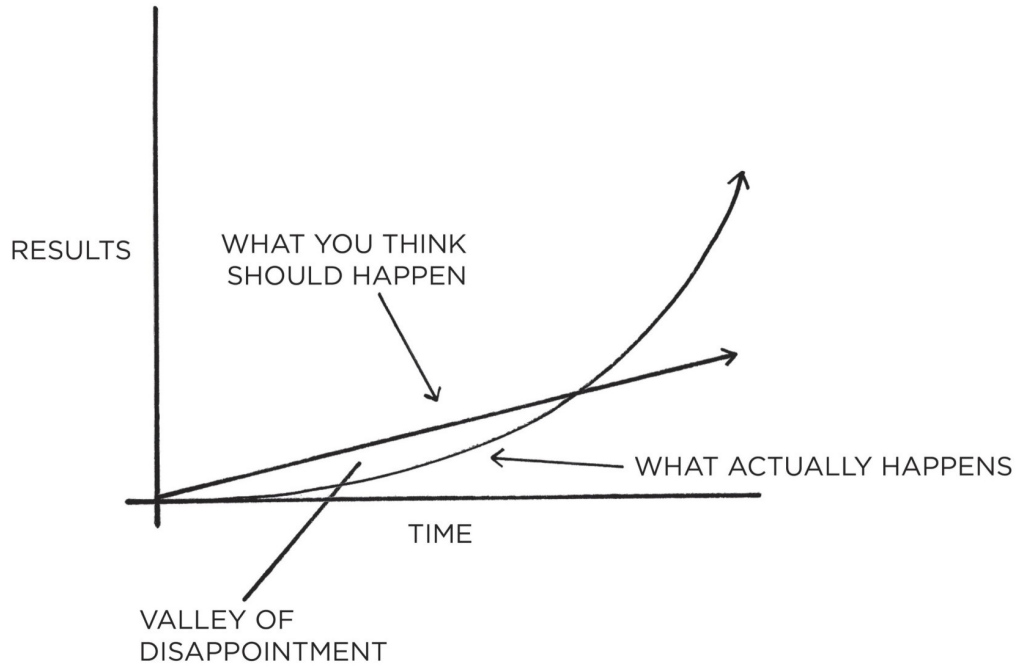


FIGURE 2: We often expect progress to be linear. At the very least, we hope it will come quickly. In reality, the results of our efforts are often delayed. It is not until months or years later that we realize the true value of the previous work we have done. This can result in a “valley of disappointment” where people feel discouraged after putting in weeks or months of hard work without experiencing any results. However, this work was not wasted. It was simply being stored. It is not until much later that the full value of previous efforts is revealed.

All big things come from small beginnings. The seed of every habit is a single, tiny decision. But as that decision is repeated, a habit sprouts and grows stronger. Roots entrench themselves and branches grow. The task of breaking a bad habit is like uprooting a powerful oak within us. And the task of building a good habit is like cultivating a delicate flower one day at a time.

But what determines whether we stick with a habit long enough to survive the Plateau of Latent Potential and break through to the other side? What is it that

causes some people to slide into unwanted habits and enables others to enjoy the compounding effects of good ones?

FORGET ABOUT GOALS, FOCUS ON SYSTEMS INSTEAD

Prevailing wisdom claims that the best way to achieve what we want in life—getting into better shape, building a successful business, relaxing more and worrying less, spending more time with friends and family—is to set specific, actionable goals.

For many years, this was how I approached my habits, too. Each one was a goal to be reached. I set goals for the grades I wanted to get in school, for the weights I wanted to lift in the gym, for the profits I wanted to earn in business. I succeeded at a few, but I failed at a lot of them. Eventually, I began to realize that my results had very little to do with the goals I set and nearly everything to do with the systems I followed.

What's the difference between systems and goals? It's a distinction I first learned from Scott Adams, the cartoonist behind the *Dilbert* comic. Goals are about the results you want to achieve. Systems are about the processes that lead to those results.

- If you're a coach, your goal might be to win a championship. Your system is the way you recruit players, manage your assistant coaches, and conduct practice.
- If you're an entrepreneur, your goal might be to build a million-dollar business. Your system is how you test product ideas, hire employees, and run marketing campaigns.
- If you're a musician, your goal might be to play a new piece. Your system is how often you practice, how you break down and tackle difficult measures, and your method for receiving feedback from your instructor.

Now for the interesting question: If you completely ignored your goals and focused only on your system, would you still succeed? For example, if you were a basketball coach and you ignored your goal to win a championship and focused only on what your team does at practice each day, would you still get results?

I think you would.

The goal in any sport is to finish with the best score, but it would be ridiculous to spend the whole game staring at the scoreboard. The only way to actually win is to get better each day. In the words of three-time Super Bowl winner Bill Walsh, “The score takes care of itself.” The same is true for other areas of life. If you want better results, then forget about setting goals. Focus on your system instead.

What do I mean by this? Are goals completely useless? Of course not. Goals are good for setting a direction, but systems are best for making progress. A handful of problems arise when you spend too much time thinking about your goals and not enough time designing your systems.

Problem #1: Winners and losers have the same goals.

Goal setting suffers from a serious case of survivorship bias. We concentrate on the people who end up winning—the survivors—and mistakenly assume that ambitious goals led to their success while overlooking all of the people who had the same objective but didn’t succeed.

Every Olympian wants to win a gold medal. Every candidate wants to get the job. And if successful and unsuccessful people share the same goals, then the goal cannot be what differentiates the winners from the losers. It wasn’t the *goal* of winning the Tour de France that propelled the British cyclists to the top of the sport. Presumably, they had wanted to win the race every year before—just like every other professional team. The goal had always been there. It was only when they implemented a *system* of continuous small improvements that they achieved a different outcome.

Problem #2: Achieving a goal is only a momentary change.

Imagine you have a messy room and you set a goal to clean it. If you summon the energy to tidy up, then you will have a clean room—for now. But if you maintain the same sloppy, pack-rat habits that led to a messy room in the first place, soon you’ll be looking at a new pile of clutter and hoping for another burst of motivation. You’re left chasing the same outcome because you never changed the system behind it. You treated a symptom without addressing the cause.

Achieving a goal only changes your life *for the moment*. That’s the counterintuitive thing about improvement. We think we need to change our results, but the results are not the problem. What we really need to change are the systems that cause those results. When you solve problems at the results

level, you only solve them temporarily. In order to improve for good, you need to solve problems at the systems level. Fix the inputs and the outputs will fix themselves.

Problem #3: Goals restrict your happiness.

The implicit assumption behind any goal is this: "Once I reach my goal, then I'll be happy." The problem with a goals-first mentality is that you're continually putting happiness off until the next milestone. I've slipped into this trap so many times I've lost count. For years, happiness was always something for my future self to enjoy. I promised myself that once I gained twenty pounds of muscle or after my business was featured in the *New York Times*, then I could finally relax.

Furthermore, goals create an "either-or" conflict: either you achieve your goal and are successful or you fail and you are a disappointment. You mentally box yourself into a narrow version of happiness. This is misguided. It is unlikely that your actual path through life will match the exact journey you had in mind when you set out. It makes no sense to restrict your satisfaction to one scenario when there are many paths to success.

A systems-first mentality provides the antidote. When you fall in love with the process rather than the product, you don't have to wait to give yourself permission to be happy. You can be satisfied anytime your system is running. And a system can be successful in many different forms, not just the one you first envision.

Problem #4: Goals are at odds with long-term progress.

Finally, a goal-oriented mind-set can create a "yo-yo" effect. Many runners work hard for months, but as soon as they cross the finish line, they stop training. The race is no longer there to motivate them. When all of your hard work is focused on a particular goal, what is left to push you forward after you achieve it? This is why many people find themselves reverting to their old habits after accomplishing a goal.

The purpose of setting goals is to win the game. The purpose of building systems is to continue playing the game. True long-term thinking is goal-less thinking. It's not about any single accomplishment. It is about the cycle of endless refinement and continuous improvement. Ultimately, it is your commitment to the *process* that will determine your *progress*.

A SYSTEM OF ATOMIC HABITS

If you're having trouble changing your habits, the problem isn't you. The problem is your system. Bad habits repeat themselves again and again not because you don't want to change, but because you have the wrong system for change.

You do not rise to the level of your goals. You fall to the level of your systems.

Focusing on the overall system, rather than a single goal, is one of the core themes of this book. It is also one of the deeper meanings behind the word *atomic*. By now, you've probably realized that an atomic habit refers to a tiny change, a marginal gain, a 1 percent improvement. But atomic habits are not just any old habits, however small. They are little habits that are part of a larger system. Just as atoms are the building blocks of molecules, atomic habits are the building blocks of remarkable results.

Habits are like the atoms of our lives. Each one is a fundamental unit that contributes to your overall improvement. At first, these tiny routines seem insignificant, but soon they build on each other and fuel bigger wins that multiply to a degree that far outweighs the cost of their initial investment. They are both small and mighty. This is the meaning of the phrase *atomic habits*—a regular practice or routine that is not only small and easy to do, but also the source of incredible power; a component of the system of compound growth.

Chapter Summary

- Habits are the compound interest of selfimprovement. Getting 1 percent better every day counts for a lot in the long-run.
- Habits are a double-edged sword. They can work for you or against you, which is why understanding the details is essential.
- Small changes often appear to make no difference until you cross a critical threshold. The most powerful outcomes of any compounding process are delayed. You need to be patient.
- An atomic habit is a little habit that is part of a larger system. Just as atoms are the building blocks of molecules, atomic habits are the building blocks of remarkable results.
- If you want better results, then forget about setting goals. Focus on your system instead.
- You do not rise to the level of your goals. You fall to the level of your systems.



2

How Your Habits Shape Your Identity (and Vice Versa)

WHY IS IT so easy to repeat bad habits and so hard to form good ones? Few things can have a more powerful impact on your life than improving your daily habits. And yet it is likely that this time next year you'll be doing the same thing rather than something better.

It often feels difficult to keep good habits going for more than a few days, even with sincere effort and the occasional burst of motivation. Habits like exercise, meditation, journaling, and cooking are reasonable for a day or two and then become a hassle.

However, once your habits are established, they seem to stick around forever—especially the unwanted ones. Despite our best intentions, unhealthy habits like eating junk food, watching too much television, procrastinating, and smoking can feel impossible to break.

Changing our habits is challenging for two reasons: (1) we try to change the wrong thing and (2) we try to change our habits in the wrong way. In this chapter, I'll address the first point. In the chapters that follow, I'll answer the second.

Our first mistake is that we try to change the wrong thing. To understand what I mean, consider that there are three levels at which change can occur. You can imagine them like the layers of an onion.

THREE LAYERS OF BEHAVIOR CHANGE

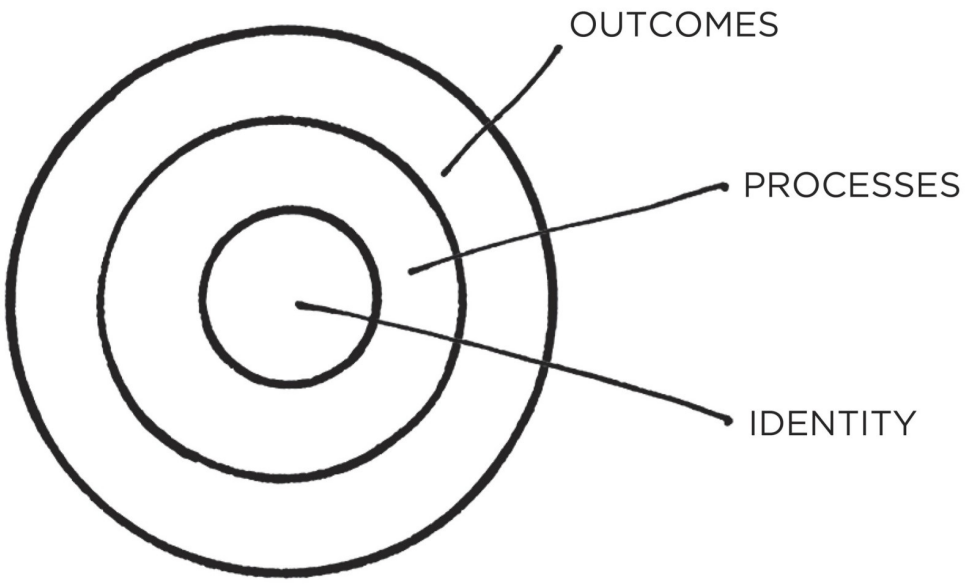


FIGURE 3: There are three layers of behavior change: a change in your outcomes, a change in your processes, or a change in your identity.

The first layer is changing your outcomes. This level is concerned with changing your results: losing weight, publishing a book, winning a championship. Most of the goals you set are associated with this level of change.

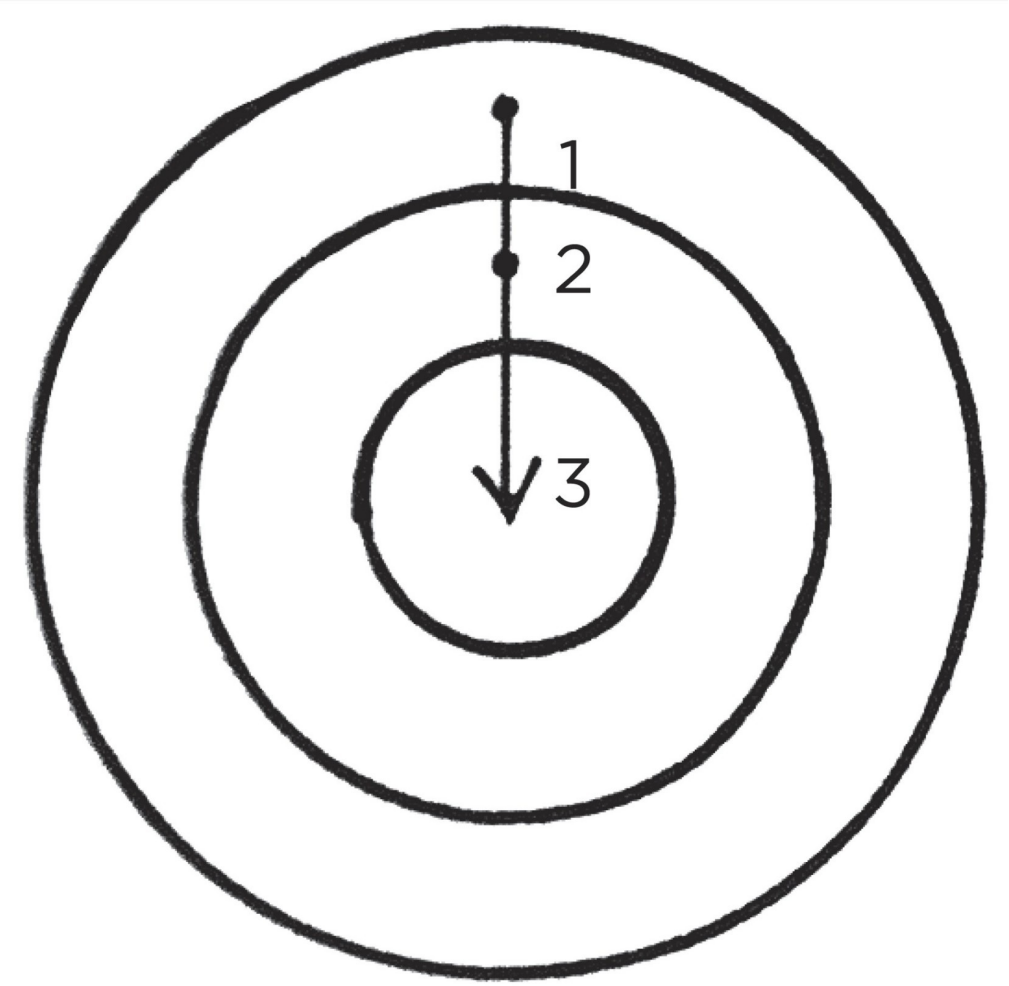
The second layer is changing your process. This level is concerned with changing your habits and systems: implementing a new routine at the gym, decluttering your desk for better workflow, developing a meditation practice. Most of the habits you build are associated with this level.

The third and deepest layer is changing your identity. This level is concerned with changing your beliefs: your worldview, your self-image, your judgments about yourself and others. Most of the beliefs, assumptions, and biases you hold are associated with this level.

Outcomes are about what you get. Processes are about what you do. Identity is about what you believe. When it comes to building habits that last—when it comes to building a system of 1 percent improvements—the problem is not that one level is "better" or "worse" than another. All levels of change are useful in their own way. The problem is the *direction* of change.

Many people begin the process of changing their habits by focusing on *what* they want to achieve. This leads us to outcome-based habits. The alternative is to build identity-based habits. With this approach, we start by focusing on *who* we wish to become.

OUTCOME-BASED HABITS



IDENTITY-BASED HABITS

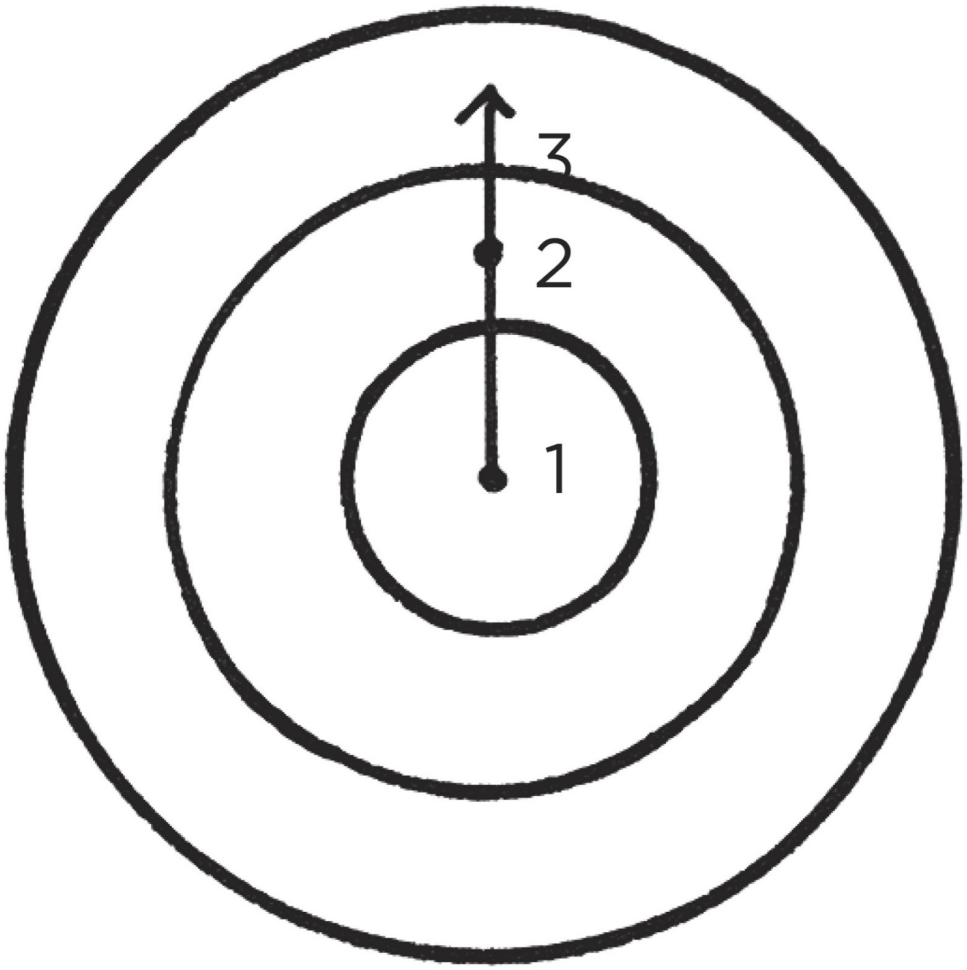


FIGURE 4: With outcome-based habits, the focus is on what you want to achieve. With identity-based habits, the focus is on who you wish to become.

Imagine two people resisting a cigarette. When offered a smoke, the first person says, “No thanks. I’m trying to quit.” It sounds like a reasonable response, but this person still believes they are a smoker who is trying to be something else. They are hoping their behavior will change while carrying around the same beliefs.

The second person declines by saying, “No thanks. I’m not a smoker.” It’s a small difference, but this statement signals a shift in identity. Smoking was part of their former life, not their current one. They no longer identify as someone who smokes.

Most people don’t even consider identity change when they set out to improve. They just think, “I want to be skinny (outcome) and if I stick to this diet, then I’ll be skinny (process).” They set goals and determine the actions they

should take to achieve those goals without considering the beliefs that drive their actions. They never shift the way they look at themselves, and they don't realize that their old identity can sabotage their new plans for change.

Behind every system of actions are a system of beliefs. The system of a democracy is founded on beliefs like freedom, majority rule, and social equality. The system of a dictatorship has a very different set of beliefs like absolute authority and strict obedience. You can imagine many ways to try to get more people to vote in a democracy, but such behavior change would never get off the ground in a dictatorship. That's not the identity of the system. Voting is a behavior that is impossible under a certain set of beliefs.

A similar pattern exists whether we are discussing individuals, organizations, or societies. There are a set of beliefs and assumptions that shape the system, an identity behind the habits.

Behavior that is incongruent with the self will not last. You may want more money, but if your identity is someone who consumes rather than creates, then you'll continue to be pulled toward spending rather than earning. You may want better health, but if you continue to prioritize comfort over accomplishment, you'll be drawn to relaxing rather than training. It's hard to change your habits if you never change the underlying beliefs that led to your past behavior. You have a new goal and a new plan, but you haven't changed *who* you are.

The story of Brian Clark, an entrepreneur from Boulder, Colorado, provides a good example. "For as long as I can remember, I've chewed my fingernails," Clark told me. "It started as a nervous habit when I was young, and then morphed into an undesirable grooming ritual. One day, I resolved to stop chewing my nails until they grew out a bit. Through mindful willpower alone, I managed to do it."

Then, Clark did something surprising.

"I asked my wife to schedule my first-ever manicure," he said. "My thought was that if I started paying to maintain my nails, I wouldn't chew them. And it worked, but not for the monetary reason. What happened was the manicure made my fingers look really nice for the first time. The manicurist even said that—other than the chewing—I had really healthy, attractive nails. Suddenly, I was proud of my fingernails. And even though that's something I had never aspired to, it made all the difference. I've never chewed my nails since; not even a single close call. And it's because I now take pride in properly caring for them."

The ultimate form of intrinsic motivation is when a habit becomes part of your identity. It's one thing to say I'm the type of person who *wants* this. It's something very different to say I'm the type of person who *is* this.

The more pride you have in a particular aspect of your identity, the more

motivated you will be to maintain the habits associated with it. If you're proud of how your hair looks, you'll develop all sorts of habits to care for and maintain it. If you're proud of the size of your biceps, you'll make sure you never skip an upper-body workout. If you're proud of the scarves you knit, you'll be more likely to spend hours knitting each week. Once your pride gets involved, you'll fight tooth and nail to maintain your habits.

True behavior change is identity change. You might start a habit because of motivation, but the only reason you'll stick with one is that it becomes part of your identity. Anyone can convince themselves to visit the gym or eat healthy once or twice, but if you don't shift the belief behind the behavior, then it is hard to stick with long-term changes. Improvements are only temporary until they become part of who you are.

- The goal is not to read a book, the goal is to *become* a reader.
- The goal is not to run a marathon, the goal is to *become* a runner.
- The goal is not to learn an instrument, the goal is to *become* a musician.

Your behaviors are usually a reflection of your identity. What you do is an indication of the type of person you believe that you are—either consciously or nonconsciously.* Research has shown that once a person believes in a particular aspect of their identity, they are more likely to act in alignment with that belief. For example, people who identified as "being a voter" were more likely to vote than those who simply claimed "voting" was an action they wanted to perform. Similarly, the person who incorporates exercise into their identity doesn't have to convince themselves to train. Doing the right thing is easy. After all, when your behavior and your identity are fully aligned, you are no longer pursuing behavior change. You are simply acting like the type of person you already believe yourself to be.

Like all aspects of habit formation, this, too, is a double-edged sword. When working for you, identity change can be a powerful force for selfimprovement. When working against you, though, identity change can be a curse. Once you have adopted an identity, it can be easy to let your allegiance to it impact your ability to change. Many people walk through life in a cognitive slumber, blindly following the norms attached to their identity.

- “I’m terrible with directions.”
- “I’m not a morning person.”
- “I’m bad at remembering people’s names.”
- “I’m always late.”
- “I’m not good with technology.”
- “I’m horrible at math.”

. . . and a thousand other variations.

When you have repeated a story to yourself for years, it is easy to slide into these mental grooves and accept them as a fact. In time, you begin to resist certain actions because “that’s not who I am.” There is internal pressure to maintain your self-image and behave in a way that is consistent with your beliefs. You find whatever way you can to avoid contradicting yourself.

The more deeply a thought or action is tied to your identity, the more difficult it is to change it. It can feel comfortable to believe what your culture believes (group identity) or to do what upholds your self-image (personal identity), even if it’s wrong. The biggest barrier to positive change at any level—individual, team, society—is identity conflict. Good habits can make rational sense, but if they conflict with your identity, you will fail to put them into action.

On any given day, you may struggle with your habits because you’re too busy or too tired or too overwhelmed or hundreds of other reasons. Over the long run, however, the real reason you fail to stick with habits is that your self-image gets in the way. This is why you can’t get too attached to one version of your identity. Progress requires unlearning. Becoming the best version of yourself requires you to continuously edit your beliefs, and to upgrade and expand your identity.

This brings us to an important question: If your beliefs and worldview play such an important role in your behavior, where do they come from in the first place? How, exactly, is your identity formed? And how can you emphasize new aspects of your identity that serve you and gradually erase the pieces that hinder you?

THE TWO-STEP PROCESS TO CHANGING YOUR IDENTITY

Your identity emerges out of your habits. You are not born with preset beliefs. Every belief, including those about yourself, is learned and conditioned through experience.*

More precisely, your habits are how you *embody* your identity. When you make your bed each day, you embody the identity of an organized person. When you write each day, you embody the identity of a creative person. When you train each day, you embody the identity of an athletic person.

The more you repeat a behavior, the more you reinforce the identity associated with that behavior. In fact, the word *identity* was originally derived from the Latin words *essentitas*, which means *being*, and *identidem*, which means *repeatedly*. Your identity is literally your "repeated beingness."

Whatever your identity is right now, you only believe it because you have proof of it. If you go to church every Sunday for twenty years, you have evidence that you are religious. If you study biology for one hour every night, you have evidence that you are studious. If you go to the gym even when it's snowing, you have evidence that you are committed to fitness. The more evidence you have for a belief, the more strongly you will believe it.

For most of my early life, I didn't consider myself a writer. If you were to ask any of my high school teachers or college professors, they would tell you I was an average writer at best: certainly not a standout. When I began my writing career, I published a new article every Monday and Thursday for the first few years. As the evidence grew, so did my identity as a writer. I didn't start out as a writer. I *became* one through my habits.

Of course, your habits are not the *only* actions that influence your identity, but by virtue of their frequency they are usually the most important ones. Each experience in life modifies your self-image, but it's unlikely you would consider yourself a soccer player because you kicked a ball once or an artist because you scribbled a picture. As you repeat these actions, however, the evidence accumulates and your self-image begins to change. The effect of one-off experiences tends to fade away while the effect of habits gets reinforced with time, which means your habits contribute most of the evidence that shapes your identity. In this way, the process of building habits is actually the process of becoming yourself.

This is a gradual evolution. We do not change by snapping our fingers and deciding to be someone entirely new. We change bit by bit, day by day, habit by habit. We are continually undergoing microevolutions of the self.

Each habit is like a suggestion: "Hey, maybe *this* is who I am." If you finish a book, then perhaps you are the type of person who likes reading. If you go to the gym, then perhaps you are the type of person who likes exercise. If you practice playing the guitar, perhaps you are the type of person who likes music.

Every action you take is a vote for the type of person you wish to become. No single instance will transform your beliefs, but as the votes build up, so does the

evidence of your new identity. This is one reason why meaningful change does not require radical change. Small habits can make a meaningful difference by providing evidence of a new identity. And if a change is meaningful, it actually is big. That's the paradox of making small improvements.

Putting this all together, you can see that habits are the path to changing your identity. The most practical way to change *who* you are is to change *what* you do.

- Each time you write a page, you are a writer.
- Each time you practice the violin, you are a musician.
- Each time you start a workout, you are an athlete.
- Each time you encourage your employees, you are a leader.

Each habit not only gets results but also teaches you something far more important: to trust yourself. You start to believe you can actually accomplish these things. When the votes mount up and the evidence begins to change, the story you tell yourself begins to change as well.

Of course, it works the opposite way, too. Every time you choose to perform a bad habit, it's a vote for that identity. The good news is that you don't need to be perfect. In any election, there are going to be votes for both sides. You don't need a unanimous vote to win an election; you just need a majority. It doesn't matter if you cast a few votes for a bad behavior or an unproductive habit. Your goal is simply to win the majority of the time.

New identities require new evidence. If you keep casting the same votes you've always cast, you're going to get the same results you've always had. If nothing changes, nothing is going to change.

It is a simple two-step process:

1. Decide the type of person you want to be.
2. Prove it to yourself with small wins.

First, decide who you want to be. This holds at any level—as an individual, as a team, as a community, as a nation. What do you want to stand for? What are your principles and values? Who do you wish to become?

These are big questions, and many people aren't sure where to begin—but they do know what kind of results they want: to get six-pack abs or to feel less

anxious or to double their salary. That's fine. Start there and work backward from the results you want to the type of person who could get those results. Ask yourself, "Who is the type of person that could get the outcome I want?" Who is the type of person that could lose forty pounds? Who is the type of person that could learn a new language? Who is the type of person that could run a successful start-up?

For example, "Who is the type of person who could write a book?" It's probably someone who is consistent and reliable. Now your focus shifts from writing a book (outcome-based) to being the type of person who is consistent and reliable (identity-based).

This process can lead to beliefs like:

- "I'm the kind of teacher who stands up for her students."
- "I'm the kind of doctor who gives each patient the time and empathy they need."
- "I'm the kind of manager who advocates for her employees."

Once you have a handle on the type of person you want to be, you can begin taking small steps to reinforce your desired identity. I have a friend who lost over 100 pounds by asking herself, "What would a healthy person do?" All day long, she would use this question as a guide. Would a healthy person walk or take a cab? Would a healthy person order a burrito or a salad? She figured if she acted like a healthy person long enough, eventually she would become that person. She was right.

The concept of identity-based habits is our first introduction to another key theme in this book: feedback loops. Your habits shape your identity, and your identity shapes your habits. It's a two-way street. The formation of all habits is a feedback loop (a concept we will explore in depth in the next chapter), but it's important to let your values, principles, and identity drive the loop rather than your results. The focus should always be on becoming that type of person, not getting a particular outcome.

THE REAL REASON HABITS MATTER

Identity change is the North Star of habit change. The remainder of this book will provide you with step-by-step instructions on how to build better habits in yourself, your family, your team, your company, and anywhere else you wish.

But the true question is: “Are you becoming the type of person you want to become?” The first step is not *what* or *how*, but *who*. You need to know who you want to be. Otherwise, your quest for change is like a boat without a rudder. And that’s why we are starting here.

You have the power to change your beliefs about yourself. Your identity is not set in stone. You have a choice in every moment. You can choose the identity you want to reinforce today with the habits you choose today. And this brings us to the deeper purpose of this book and the real reason habits matter.

Building better habits isn’t about littering your day with life hacks. It’s not about flossing one tooth each night or taking a cold shower each morning or wearing the same outfit each day. It’s not about achieving external measures of success like earning more money, losing weight, or reducing stress. Habits can help you achieve all of these things, but fundamentally they are not about *having* something. They are about *becoming* someone.

Ultimately, your habits matter because they help you become the type of person you wish to be. They are the channel through which you develop your deepest beliefs about yourself. Quite literally, you become your habits.

Chapter Summary

- There are three levels of change: outcome change, process change, and identity change.
- The most effective way to change your habits is to focus not on what you want to achieve, but on who you wish to become.
- Your identity emerges out of your habits. Every action is a vote for the type of person you wish to become.
- Becoming the best version of yourself requires you to continuously edit your beliefs, and to upgrade and expand your identity.
- The real reason habits matter is not because they can get you better results (although they can do that), but because they can change your beliefs about yourself.



3

How to Build Better Habits in 4 Simple Steps

IN 1898, A psychologist named Edward Thorndike conducted an experiment that would lay the foundation for our understanding of how habits form and the rules that guide our behavior. Thorndike was interested in studying the behavior of animals, and he started by working with cats.

He would place each cat inside a device known as a puzzle box. The box was designed so that the cat could escape through a door "by some simple act, such as pulling at a loop of cord, pressing a lever, or stepping on a platform." For example, one box contained a lever that, when pressed, would open a door on the side of the box. Once the door had been opened, the cat could dart out and run over to a bowl of food.

Most cats wanted to escape as soon as they were placed inside the box. They would poke their nose into the corners, stick their paws through openings, and claw at loose objects. After a few minutes of exploration, the cats would happen to press the magical lever, the door would open, and they would escape.

Thorndike tracked the behavior of each cat across many trials. In the beginning, the animals moved around the box at random. But as soon as the lever had been pressed and the door opened, the process of learning began. Gradually, each cat learned to associate the action of pressing the lever with the reward of escaping the box and getting to the food.

After twenty to thirty trials, this behavior became so automatic and habitual that the cat could escape within a few seconds. For example, Thorndike noted, "Cat 12 took the following times to perform the act. 160 seconds, 30 seconds, 90 seconds, 60, 15, 28, 20, 30, 22, 11, 15, 20, 12, 10, 14, 10, 8, 8, 5, 10, 8, 6, 6, 7."

During the first three trials, the cat escaped in an average of 1.5 minutes. During the last three trials, it escaped in an average of 6.3 seconds. With practice, each cat made fewer errors and their actions became quicker and more automatic. Rather than repeat the same mistakes, the cat began to cut straight to

the solution.

From his studies, Thorndike described the learning process by stating, "behaviors followed by satisfying consequences tend to be repeated and those that produce unpleasant consequences are less likely to be repeated." His work provides the perfect starting point for discussing how habits form in our own lives. It also provides answers to some fundamental questions like: What are habits? And why does the brain bother building them at all?

WHY YOUR BRAIN BUILDS HABITS

A habit is a behavior that has been repeated enough times to become automatic. The process of habit formation begins with trial and error. Whenever you encounter a new situation in life, your brain has to make a decision. *How do I respond to this?* The first time you come across a problem, you're not sure how to solve it. Like Thorndike's cat, you're just trying things out to see what works.

Neurological activity in the brain is high during this period. You are carefully analyzing the situation and making conscious decisions about how to act. You're taking in tons of new information and trying to make sense of it all. The brain is busy learning the most effective course of action.

Occasionally, like a cat pressing on a lever, you stumble across a solution. You're feeling anxious, and you discover that going for a run calms you down. You're mentally exhausted from a long day of work, and you learn that playing video games relaxes you. You're exploring, exploring, exploring, and then—BAM—a reward.

After you stumble upon an unexpected reward, you alter your strategy for next time. Your brain immediately begins to catalog the events that preceded the reward. *Wait a minute—that felt good. What did I do right before that?*

This is the feedback loop behind all human behavior: try, fail, learn, try differently. With practice, the useless movements fade away and the useful actions get reinforced. That's a habit forming.

Whenever you face a problem repeatedly, your brain begins to automate the process of solving it. Your habits are just a series of automatic solutions that solve the problems and stresses you face regularly. As behavioral scientist Jason Hreha writes, "Habits are, simply, reliable solutions to recurring problems in our environment."

As habits are created, the level of activity in the brain *decreases*. You learn to lock in on the cues that predict success and tune out everything else. When a similar situation arises in the future, you know exactly what to look for. There is

no longer a need to analyze every angle of a situation. Your brain skips the process of trial and error and creates a mental rule: if this, then that. These cognitive scripts can be followed automatically whenever the situation is appropriate. Now, whenever you feel stressed, you get the itch to run. As soon as you walk in the door from work, you grab the video game controller. A choice that once required effort is now automatic. A habit has been created.

Habits are mental shortcuts learned from experience. In a sense, a habit is just a memory of the steps you previously followed to solve a problem in the past. Whenever the conditions are right, you can draw on this memory and automatically apply the same solution. The primary reason the brain remembers the past is to better predict what will work in the future.

Habit formation is incredibly useful because the conscious mind is the bottleneck of the brain. It can only pay attention to one problem at a time. As a result, your brain is always working to preserve your conscious attention for whatever task is most essential. Whenever possible, the conscious mind likes to pawn off tasks to the nonconscious mind to do automatically. This is precisely what happens when a habit is formed. Habits reduce cognitive load and free up mental capacity, so you can allocate your attention to other tasks.

Despite their efficiency, some people still wonder about the benefits of habits. The argument goes like this: “Will habits make my life dull? I don’t want to pigeonhole myself into a lifestyle I don’t enjoy. Doesn’t so much routine take away the vibrancy and spontaneity of life?” Hardly. Such questions set up a false dichotomy. They make you think that you have to choose between building habits and attaining freedom. In reality, the two complement each other.

Habits do not restrict freedom. They create it. In fact, the people who don’t have their habits handled are often the ones with the *least* amount of freedom. Without good financial habits, you will always be struggling for the next dollar. Without good health habits, you will always seem to be short on energy. Without good learning habits, you will always feel like you’re behind the curve. If you’re always being forced to make decisions about simple tasks—when should I work out, where do I go to write, when do I pay the bills—then you have less time for freedom. It’s only by making the fundamentals of life easier that you can create the mental space needed for free thinking and creativity.

Conversely, when you have your habits dialed in and the basics of life are handled and done, your mind is free to focus on new challenges and master the next set of problems. Building habits in the present allows you to do more of what you want in the future.